SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	NOVA
Project / Product Name	EventFlow AI
Final Live Deployment URL	https://event-flow-ai-customer-portal.replit.app
Codebase Repository URL	https://github.com/Joycemsan328/SDC_Hackathon.git
Name (Leader)	Tan Le Han
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2. Project Description

[Maximum 250 words]

EventFlow AI is a system that helps SMEs to organise, manage, and plan the whole event planning process in one platform. *EventFlow AI* can help to solve the challenges that SMEs face currently, such as manual redundant and repetitive work, time consuming and fragmented event planning flow.

We proposed a solution flow to solve the challenges faced by SMEs. Customers can submit the event requirements according to their preferences after they signed in to *EventFlow AI*. AI will then generate two event proposals based on customer's requirements and send them to the business' page. AI then helps SMEs on designing solutions manually. By applying computer algorithms and data from web sources, AI will generate the most ideal solution to help SMEs. Businesses and professionals will review the proposal prepared by AI, and will make the final decision. After getting the approval, customers can review and compare both proposals, then confirm the final proposal to work on. Customers can still change and resubmit their event request after reviewing the approved proposal. The system will update automatically, making it easier for SMEs.

EventFlow AI maintains a shared AI engine and database infrastructure while adapting its data input fields, prompt templates and operational requirements to different industries. This allows *EventFlow AI* to achieve horizontal scalability and expand beyond event management SMEs into other industries.

EventFlow AI aims to reduce repetitive work on proposal preparation and ease the manual workload, providing SMEs more room to connect and communicate with customers.

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

EventFlow AI is an AI-powered event planning platform designed to streamline the process of event proposal generation, business review, and customer decision-making. The system integrates modern frontend and backend technologies, cloud-based database services, and Generative AI into a unified workflow.

1. Frontend Architecture – React, TypeScript and Vite

The application is developed using React, TypeScript, and Vite, providing a responsive and interactive web interface with two main portals:

- **Customer Portal:** Allows customers to register, log in, submit event requirements, view AI-generated proposals, select their preferred plan, and request modifications.
- **Business Portal:** Enables business users to manage event requests, review AI-generated proposals, approve or request changes to proposals, and monitor event planning activities.

Both portals communicate and connect through the backend and database using Supabase to store, retrieve and update relevant information.

2. Backend Architecture – Node.js and Express.js

The backend is developed using Node.js and Express.js, serving as the application layer connecting the frontend, database, and AI generation service. It helps with:

- Processing customer event requests.
- Validating user authentication and authorization.
- Managing AI proposal generation requests.
- Communicating with the RunPod AI inference endpoint.
- Processing and storing proposal data.
- Supporting the business approval and customer response workflow.

3. Generative AI Integration – RunPod and Qwen3-8B

EventFlow AI integrates the Qwen3-8B large language model through RunPod's cloud-based AI inference infrastructure.

When a customer submits event requirements, the backend prepares a structured AI prompt containing information such as event type, date, location, budget, number of guests, theme, and required services.

The AI model generates two alternative event proposals, Plan A and Plan B, with different concepts and budget allocations. Each proposal can include event descriptions, service recommendations, timelines, estimated costs, and budget breakdowns.

The generated proposals are processed by the backend and stored in the database before being presented to business users for review.

This human-in-the-loop approach ensures that AI-generated recommendations undergo business evaluation before customers receive them.

4. Database and Authentication – Supabase

Supabase provides PostgreSQL database services and user authentication.

The database manages essential application information, including:

- User profiles and account roles.
- Customer event requests and requirements.
- AI-generated event proposals.
- Business approval decisions and proposal statuses.
- Customer plan selections and change requests.

Supabase Authentication manages user sign-in and registration, while Row Level Security (RLS) policies help enforce role-based access and protect customer information.

5. Deployment and Hosting – Replit

EventFlow AI is hosted and deployed using Replit, providing an accessible cloud-based environment for the application.

Replit supports the deployment of the frontend and backend, allowing customers and business users to access the platform through a web browser.

Sensitive configuration values, including database credentials and AI API keys, are managed through environment variables and deployment secrets

6. Integrated System Workflow

The overall system operates through the following process:

1. A customer registers or logs in and submits event requirements through the Customer Portal.
2. The application stores the event request in Supabase.
3. The backend processes the request and sends structured event requirements to the Qwen3-8B model hosted on RunPod.

4. The AI generates two personalized event proposals, Plan A and Plan B.
5. The backend validates and stores the generated proposals in Supabase.
6. Business users review the proposals through the Business Portal and approve them before customer access.
7. Once approved, customers can compare both proposals, select their preferred option, or submit a change request.
8. Customer responses are stored in Supabase and made available to the Business Portal for subsequent processing.

By integrating React, TypeScript, Express.js, Supabase, RunPod AI, and Replit, *EventFlow AI* delivers a centralized event planning solution that combines automated proposal generation with human oversight, structured data management, and an accessible web-based user experience.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

During the 48-hour physical hackathon, our team successfully developed, integrated, tested and deployed *EventFlow AI*, a full-stack AI-powered event planning platform that connects customers and event businesses through an intelligent proposal generation and approval workflow.

Key Implementations and Deliverables

We developed two interconnected web portals using React, TypeScript, and Vite. The Customer Portal enables users to register, submit event requirements, review personalized proposals, select preferred plans, and submit change requests. The Business Portal allows event businesses to manage customer requests, review AI-generated proposals, approve plans, and monitor event planning activities.

For backend development, we implemented Node.js and Express.js, integrated with Supabase PostgreSQL for data storage, authentication, and role-based access control. We also integrated Qwen3-8B through RunPod, enabling the generation of two customized event proposals (Plan A and Plan B), including event concepts, timelines, service recommendations, and estimated budget breakdowns.

The application was deployed through Replit, providing a publicly accessible working prototype.

Technical Challenges and Improvements

During development, our team encountered several integration challenges, including frontend and backend compatibility issues, database schema mismatches, authentication restrictions, and AI inference costs.

To address these issues, we consolidated development branches, resolved database integration errors, strengthened role-based authorization, and introduced controlled AI generation settings to prevent unnecessary API expenditure.

Compared with our Phase 1 preliminary design, we prioritized a more practical and reliable human-in-the-loop workflow, ensuring that AI-generated proposals undergo business review before reaching customers.

Final Outcome

We delivered a functional, cloud-deployed prototype demonstrating customer request submission, AI-assisted proposal generation, business approval, and customer decision-making, supported by an integrated database and backend infrastructure.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

The data and evidence you provide in this section will be used to assess ***SDG Impact, Technical Execution and Innovation scores.*** *Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	Your Response
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	The prototype aims to improve operational efficiency, supporting the 33% of SMBs that expect technology to increase operational efficiency.

Metric Category	Your Response
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	Our evidence is through the research from *Microsoft study: Small and medium-sized businesses growing fast by embracing digital technologies*. (2022, September 8). The Official Microsoft Blog. https://blogs.microsoft.com/blog/2022/09/08/microsoft-study-small-and-medium-sized-businesses-growing-fast-by-embracing-digital-technologies/
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	*EventFlow AI* aims to create socio-economic value by making event planning more accessible, efficient, and inclusive for both customers and SMEs. The system supports SMEs by reducing administrative workloads through AI-powered planning tools and centralised event management. This helps businesses with limited manpower improve efficiency and focus on growth. Beyond commercial use, *EventFlow AI* can assist schools, NGOs, and community organisations in planning events with limited budgets and resources.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Our website is designed as a scalable AI-powered planning platform that can be adapted to different SME industries, including retail, food and beverage, and marketing services. Future development could include affordable subscription plans and integrations with existing business management systems, making the solution more accessible to SMEs.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

EventFlow AI aims to reduce the time required for event proposal preparation by integrating Generative AI into the planning workflow. Instead of manually preparing multiple event proposals, the system uses the Qwen3-8B model deployed through RunPod to generate two personalized proposals (Plan A and Plan B) based on customer requirements.

Average AI Proposal Generation Time

$$T_{AI,avg} = \frac{\sum_{i=1}^{n} T_i}{n}$$

Where:

- T_i = Time taken to complete the i-th AI proposal generation.
- n = Total number of successful tests.
- $T_{AI,avg}$ = Average AI generation time.

Efficiency Improvement

$$E = \frac{T_{Manual} - T_{AI}}{T_{Manual}} \times 100\,\%$$

By automating the initial proposal drafting process, *EventFlow AI* has the potential to reduce repetitive planning tasks and accelerate proposal preparation. The integration of RunPod AI, Supabase, and the Business Portal demonstrates the technical feasibility of an AI-assisted, human-reviewed event planning workflow.

The system also incorporates controlled AI generation quotas to manage inference costs, supporting its operational feasibility.

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

With access to corporate funding, industry grants, and partnership opportunities, our team aims to transform the current prototype into a scalable AI-powered event planning platform.

First, we would invest in cloud infrastructure, database architecture, and AI integration to improve system performance, data storage, and processing capacity as the number of users grows. This would allow the system to handle increasing requests from customers and SMEs efficiently.

Next, we would collaborate with local SMEs and event service providers through SDC’s Rotary network to collect real-world data, including service pricing, availability, customer preferences, and event requirements. These partnerships would help improve the accuracy and relevance of our AI-generated plans while expanding our user network.

We would also conduct pilot testing with selected customers and SMEs to evaluate the platform’s usability, AI performance, and overall market demand. User usage data would guide continuous improvements, while appropriate data security and privacy measures would ensure responsible data management.

In the long term, we aim to expand our partnerships and users across Malaysia before exploring regional markets. By continuously improving our AI capabilities and building a strong network of businesses, we hope to establish a reliable and scalable platform that simplifies event planning for both customers and businesses.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

- ☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.
- ☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.
- ☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.
- ☑ The final pitch deck has been exported and saved as a flattened PDF file.
- ☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No ☑ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.

				
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31/10/2026